

Vizualization and Analysis of Sexual Assault Prosecution in the Chicago Area

Abigail Kratman & Mikayla Prue, Spring 2026

Overview and Motivation

Provide an overview of the project goals and the motivation for it. Consider that this will be read by people who did not see your project proposal.

This project began by working to find a common interest amongst team members, focusing on topics we were both keen to do in-depth research into. We landed originally on the abandonment of sexual assault kits (colloquially “rape kits”). Leaving these kits untested means that rapists throughout the United States go uncaught and unpersecuted every single day, adding even more numbers to the already staggering number of rapists who are never punished for their crimes. It is already terrifying– and often very difficult logistically– for the victims of rape to report or ask for help; leaving untested kits means even those who are able to report may never receive justice.

However, this original topic had one very large problem: there simply isn’t enough clear, publicly gathered data to make a substantial project. This topic has not been explored to the depth that the team wanted to work with. Thus, we pivoted into a similar topic with more accessible data points: assault in Chicago, which we then filtered further into crafting a dataset entirely based on criminal sexual assault charges in Chicago throughout 2001 to 2004. This topic allowed us to focus on our driving interest– a data set that could truly be implemented to support the study and support of victims of crime.

Related Works

We decided that using locational markers would be the best way to visualize this data set, seeing as the real point of interest within it comes from the population and locations of the crimes. In order to do this, we first researched how crime data has traditionally been modeled. We hit the books and found an article^[1] which gave us the idea to use literal maps and plot our data on them, in order to make the data feel less like numbers on a page. Our primary visualization-based interest is how to portray violence in data in a way that is striking and encompasses the real scale of harm. We felt that adding the data to a map of the area the crime

was actually committed in would help with that— realizing every data point was someone’s nightmare really made the work feel grounded and meaningful.

In order to make the maps, the team researched different R packages which would allow us to take the data we already have and plot it onto a map of the Chicago area. We researched the leaflet package and studied multiple articles^{[2][3]} to understand how to use it, and this allowed us to begin work in earnest.

Lastly, the team studied former criminal activity research reports, including the one found in the IEEE-published research paper *Big Data Analytics and Mining for Effective Visualization and Trends Forecasting of Crime Data*^[4]. This paper inspired the cluster implementation seen later in the report, as well as the LLM experiments, and was generally a very helpful guide. Other research papers, especially A Scalable Architecture for Visualizing and Analyzing Crime Data in LA^[5], were also used for initial brainstorming.

Questions of Interest

What questions are you trying to answer? How did these questions evolve over the course of the project? What new questions did you consider in the course of your analysis?

The question we were interested in exploring was **how to showcase incidents of crime** in a way which encompasses scale and the **genuinely disturbing commonality of assault**. We also were interested in a simpler question; how do the assaults committed in Chicago fall along population lines?

We settled into our primary research question. How could we visualize what violence means, and how it functions in a big city? Through this line of thinking, we came to our final decisions on how we wanted to visualize, and where we could go the extra mile.

Data

We sourced our data from Kaggle^[6]: an extensive set that catalogs all crimes in Chicago from 2001-2004. This was initially a massive dataset with 1,923,615 crimes with 23 different variables such as type of crime and exact location of the crime. As we quickly found, this was simply far too expansive to do meaningful work with: we had to narrow it down.

We decided to go back to our original focus and specifically look at the data which reported criminal sexual assaults. We cleaned the data set, removing any null values or any crimes that did not list the necessary location data needed for mapping. This left us with a data set of 6,554 crimes. We combined this data with a 2000 census of Chicago with help from the library tinycensus, so that we could explore which census tract holds the highest rate of crime. This allowed us to go the extra mile in terms of advanced analysis, rather than just focusing on visualizing a single dataset.

Exploratory Data Analysis

What visualizations did you use to initially look at your data? What insights did you gain? How did these insights inform your design?

Our initial visualizations started with splitting the data along the pre-existing categories: for example, a map splitting the crimes between daytime and nighttime. While this was valuable— they looked visually compelling and allowed us to practice the actual implementation of the data— we wanted to apply a secondary layer of analysis.

Thus, we took the data we had and compared it to census data from that time using the `tidycensus` library, creating a choropleth map that displayed the rate of crime per 100 citizens throughout each census tract in Chicago. This helped us to put into perspective how many crimes were being committed against a clear benchmark (the population of each census track). To support this “main attraction”, we also created visualizations such as a cluster map of the crimes committed, a heat map, a trend tracker of assaults, a “circadian rhythm” studying how often crimes occur throughout the day, and a categorical analyzer of the crime data.

Taking these visualizations, we took our work even further, creating a second choropleth map which studied the *severity* of crimes in each census track (with aggravated assault weighing heavier than non-aggravated). Both of these choropleth maps were invaluable to study how the violence of these crimes fell across the city.

Then, of course, we focused on making these designs visually appealing. This involved the usage of the libraries (such as `Leaflet`) to pick appealing and accessible color palettes, as well as design the UI. This is discussed more in the section below.

Design Evolution

What are the different visualizations you considered? Justify the design decisions you made using the perceptual and design principles you learned in the course. Did you deviate from your proposal?

There were many paths we could have taken for visualizing this project, but after some debate, we decided on entirely geographical-based maps, which allowed for clearer analysis of how Chicago's crime rate changes throughout the city lines. It became clear that the heat map and severity map were solid ways to visualize this change, while the time of day map was interesting visually but didn't offer as many clear clues towards the location's effect on assault rates. We kept our initial heat map, and evolved the severity map into the choropleth map discussed in the previous section. The severity map was especially useful with clear backing research: aggravated assault, a felony, is far more severe than non-aggravated (simply a misdemeanor). Seeing the differences between them in each area allowed for an understanding of the severity of violence.

Meanwhile, we also employed the trend graph for understanding how the rate of assault cases changed over the years studied. While it's a simple visualization, it's employed often for a reason: it is startling to register how many fewer cases were pursued annually from 2001 to 2004. We added in a second basic map tracking the hours throughout which crimes are committed, in order to use the time of day values for something more compelling on a research level (as one might guess, the hour with the highest crime is in fact midnight).

Lastly, our crowning achievement (with thanks for the feedback to add this change!) was our Population vs. Crime Rate Analysis choropleth graph, which went through multiple iterations. This graph compares each census tract to the rate of crime it experiences, making a gradient of crime-per-populace. It's visualized in chunks based on the census tracts so that users can view the actual number of crimes committed throughout the populace – to use a colloquial phrase, to showcase to users that *it could happen to you*.

As discussed in the overview section, we deviated greatly from our original proposal. However, this deviation was valuable: we employed a much stronger dataset and were able to delve into locational visualizations, which neither team member had large amounts of experience with. Employing the basic principles of data visualization-- showcasing accurate, clear, consistent data-- was the key to making this process a success, and we did so with a multitude of visualization methods.

Implementation

In order to implement this project we had to take the data and create visualizations which allowed for pattern recognition and exploration of the initially static data we were working with. To do this, we first created maps using the leaflet package in R. We started with basic point level mapping, which used `addCircleMarkers` to create data nodes. These data nodes (when clicked) show the date, description, and location of the crime, which allowed users to see the data mapped out and gain further information about the crime when investigating further.

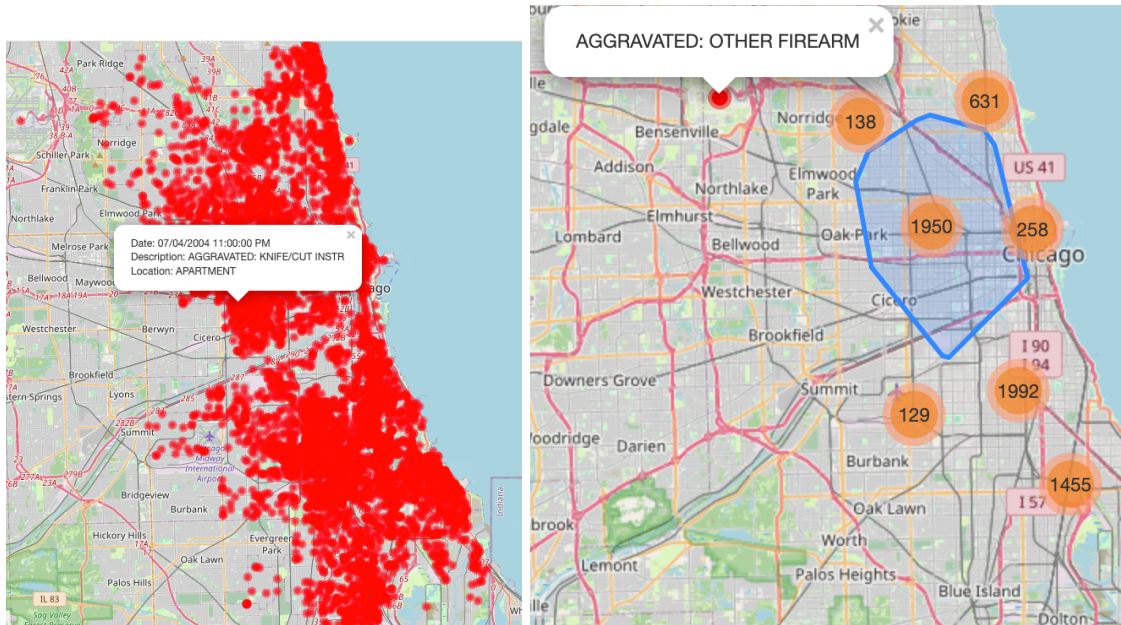


Fig 1 (a,b)) Original Vis & Final Clustered Vis of All Crimes

To address the overlapping markers in higher crime areas, we added the heatmap, as it uses continuous color gradients to show density. By zooming in or out on the heat map the user can see the data as zones rather than the individual points. When more zoomed out, the heatmap does not give a lot of information but the more specific area is focused in on it is easier to see the areas of high occurrence of sexual assault that are not as easily seen in the original point mapping.

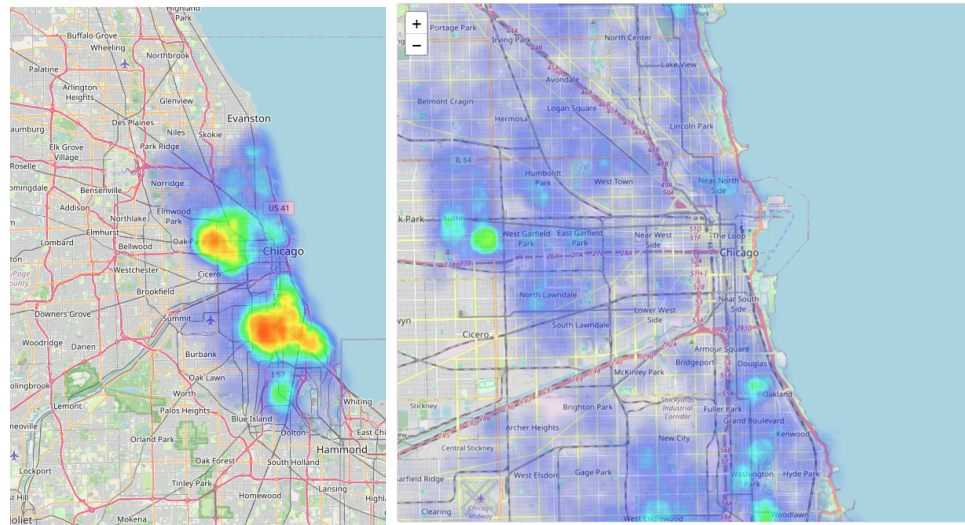


Fig 2 (a,b)) Heat Map (Zoomed Out & In)

We also wanted to correct population biases where higher density areas would appear to have more sexual assault. We did this using the tidycensus API, and integrated that information to our crime data. We took the geometric data provided within our original data set and converted it to spatial objects which allowed us to use the same coordinate system as the census data. This allowed us to aggregate the data by tract where they could be normalized by a rate of per 100 residents. We used the Viridis color scale as a color blind friendly way of showing which areas had the higher amount of assaults even when accounting for population. We included a pop up that allows the user to click individual areas and it will indicate the assaults instances per 100 in case the coloring is not readable in a specific area. We then built on this map to create a severity counter map based on the same principles.

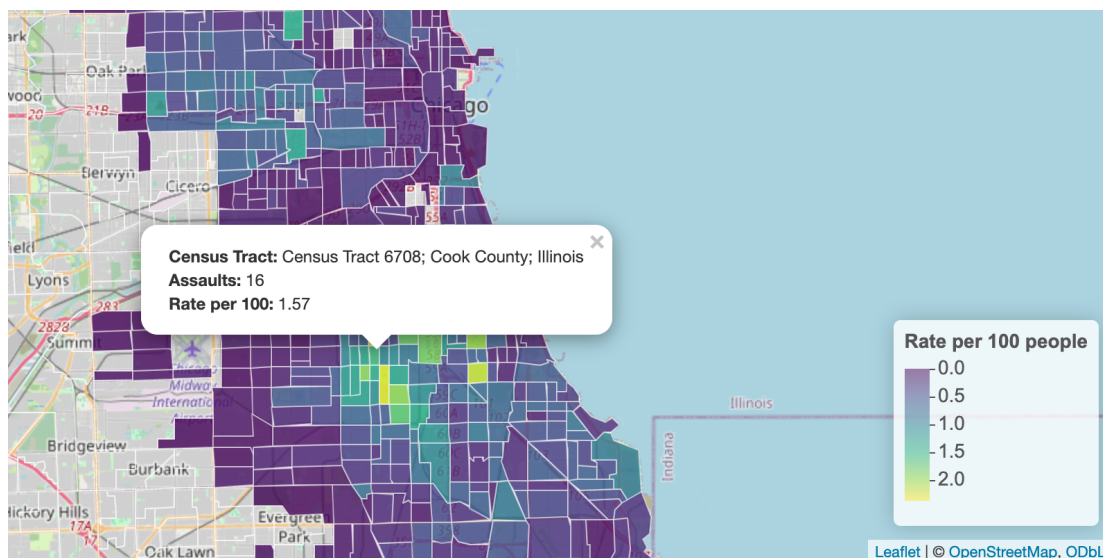


Fig 3) Population Crime Rate Analysis Map

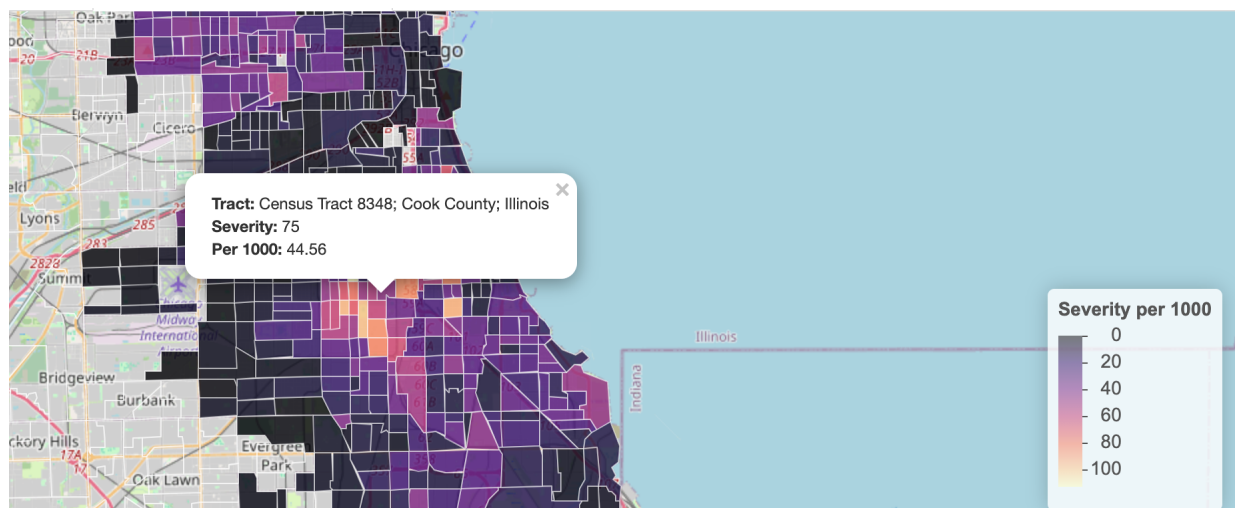


Fig 4) Population Crime Rate by Severity Analysis Map

As an added bonus— beyond just the leaflet package, we wanted to break down some of the data that spans from 2001 to 2004. We decided to look specifically at the arrest data and used ggplot2 to look at arrest rates from 2001 to 2004, as well as the rate of crimes throughout the day.

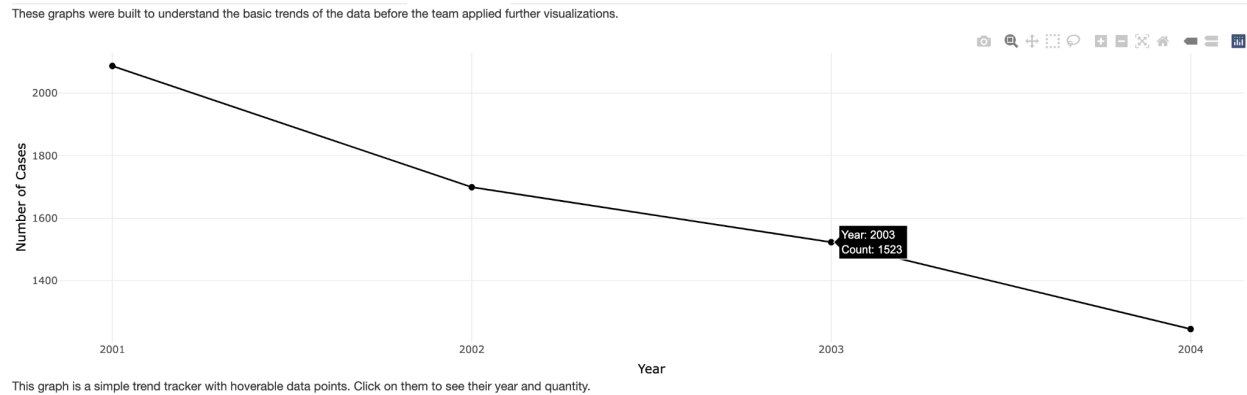


Fig 5) Arrest Trend Visualization

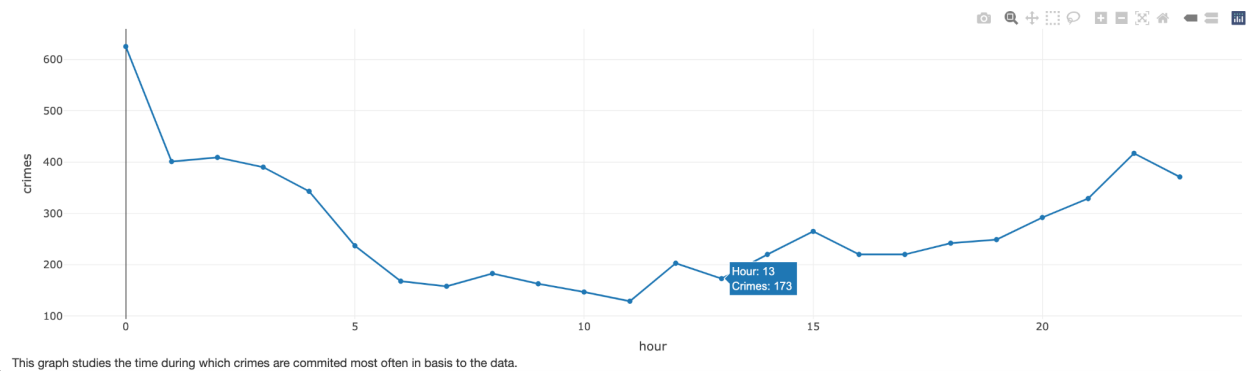


Fig 6) Hourly Visualization

Lastly, as an added study method, we implemented an LLM on our team’s local machine which analyzed the given data with the following prompt: “You are a professional crime data analyst. Analyze the yearly and hourly trends and find a conclusion.” To do this, we used Ollama’s Mistral LLM. It only runs locally, but was an interesting challenge.

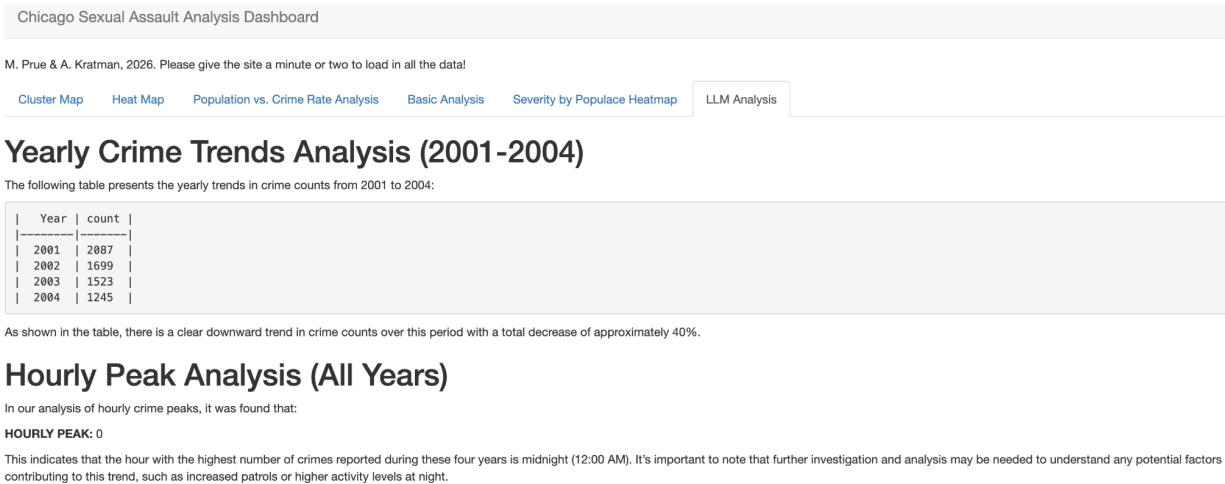


Fig 6) LLM results

Evaluation & Conclusions

What did you learn about the data by using your visualizations? How did you answer your questions? How well does your visualization work, and how could you further improve it?

The primary things we learned on this project was a deep dive into visualizing locational data— a far cry from the usual computer science instincts of scatterplots or MATLAB's bread and butter. Mapping makes the data significantly more tangible, though in trade maps are harder to work with; data can easily overlap and become harder to interpret. We answered our questions about how to interpret the data in the most interesting/engaging way (especially in regards to how to showcase the widespread violence of assault crime) by investigating other variables, such as specific location types and the times assault occurs. Overall, our visualizations functioned well and we were able to see variables that had previously been ignored.

Showcasing data in different ways helps signify its usefulness. For example, mapping it out can help law enforcement see trends in the data that are harder to pick up on by just looking it over or running averages. Data like this is often ignored because it's seen as too difficult and heavy, or else not worth pursuing due to institutional victim blaming bias; it's a flaw in our law enforcement systems. We collect this data; it should be used to help the public understand crime and its presence in our communities. It is simply appalling how many crimes go unseen because there isn't enough care taken into collecting and preserving the data— one of our first steps in the process ended up being eliminating hundreds of data points that were not collected properly and thus rendered unusable. Overall, this project was a fascinating lesson into visualizations and the work that must be done to showcase real-world events in an engaging and educational matter, answering our primary question of interest.

Bibliography

“3 Wrangling Data – Learn Crime Mapping with R.” *Lesscrime.info*, 2024,

books.lesscrime.info/learncrimemapping/03_data_wrangling/.

“Earth Analytics.” *Earth Data Science - Earth Lab*, 5 Apr. 2017,

earthdatascience.org/courses/earth-analytics/get-data-using-apis/leaflet-r/.

“Big Data Analytics and Mining for Effective Visualization and Trends Forecasting of Crime Data.” 2026.

[ieeexplore-ieee-org.ezpv7-web-p-u01.wpi.edu/stamp/stamp.jsp?tp=&arnumber=8768367
&tag=1](https://ieeexplore-ieee-org.ezpv7-web-p-u01.wpi.edu/stamp/stamp.jsp?tp=&arnumber=8768367&tag=1).

“A Scalable Architecture for Visualizing and Analyzing Crime Data in LA.” *Harrisburg*

University of Science and Technology, 1 Mar. 2025,

digitalcommons.harrisburgu.edu/cgi/viewcontent.cgi?article=1015&context=other-works.

“Crimes in Chicago.” *Kaggle*, www.kaggle.com/datasets/currie32/crimes-in-chicago.